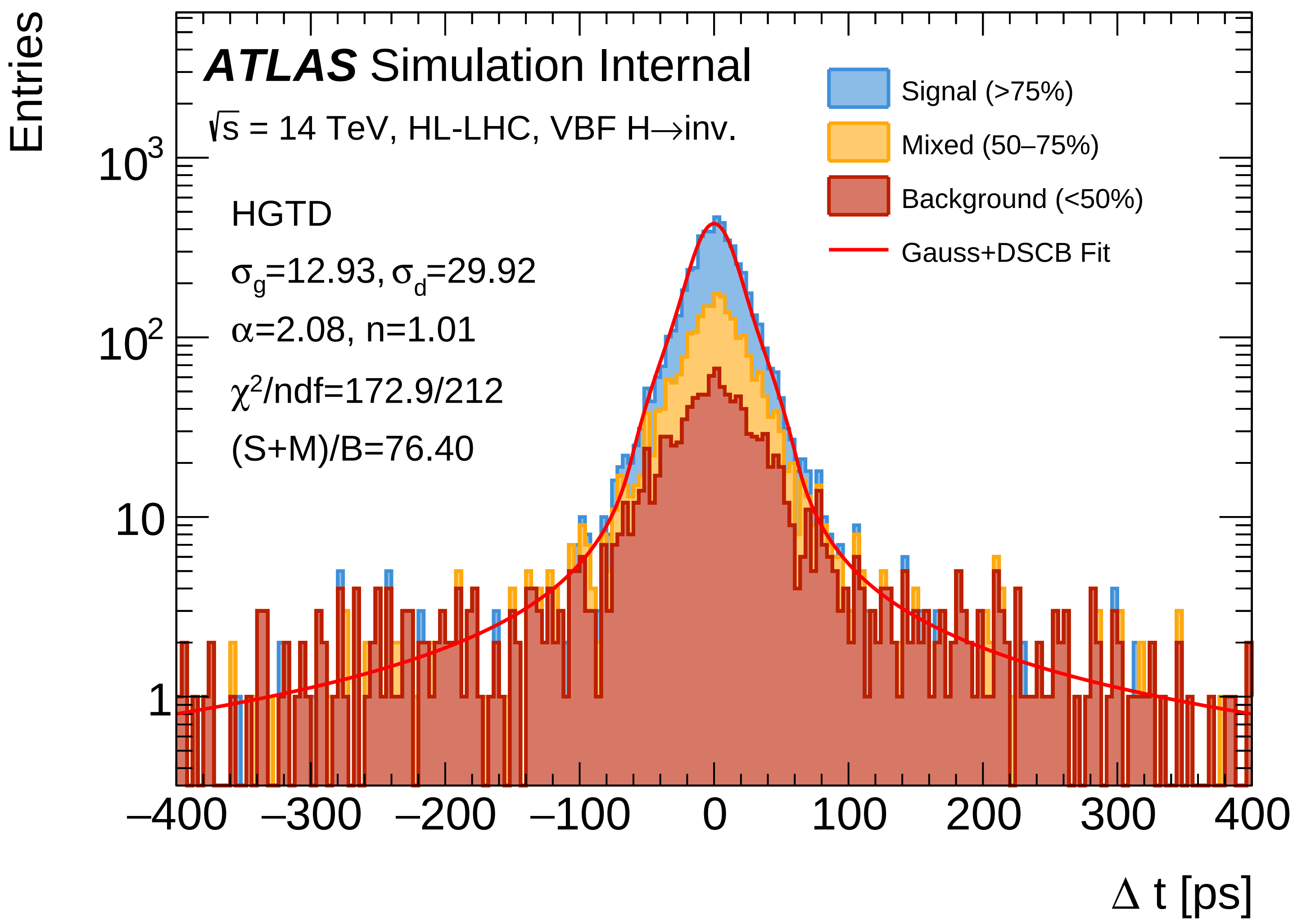
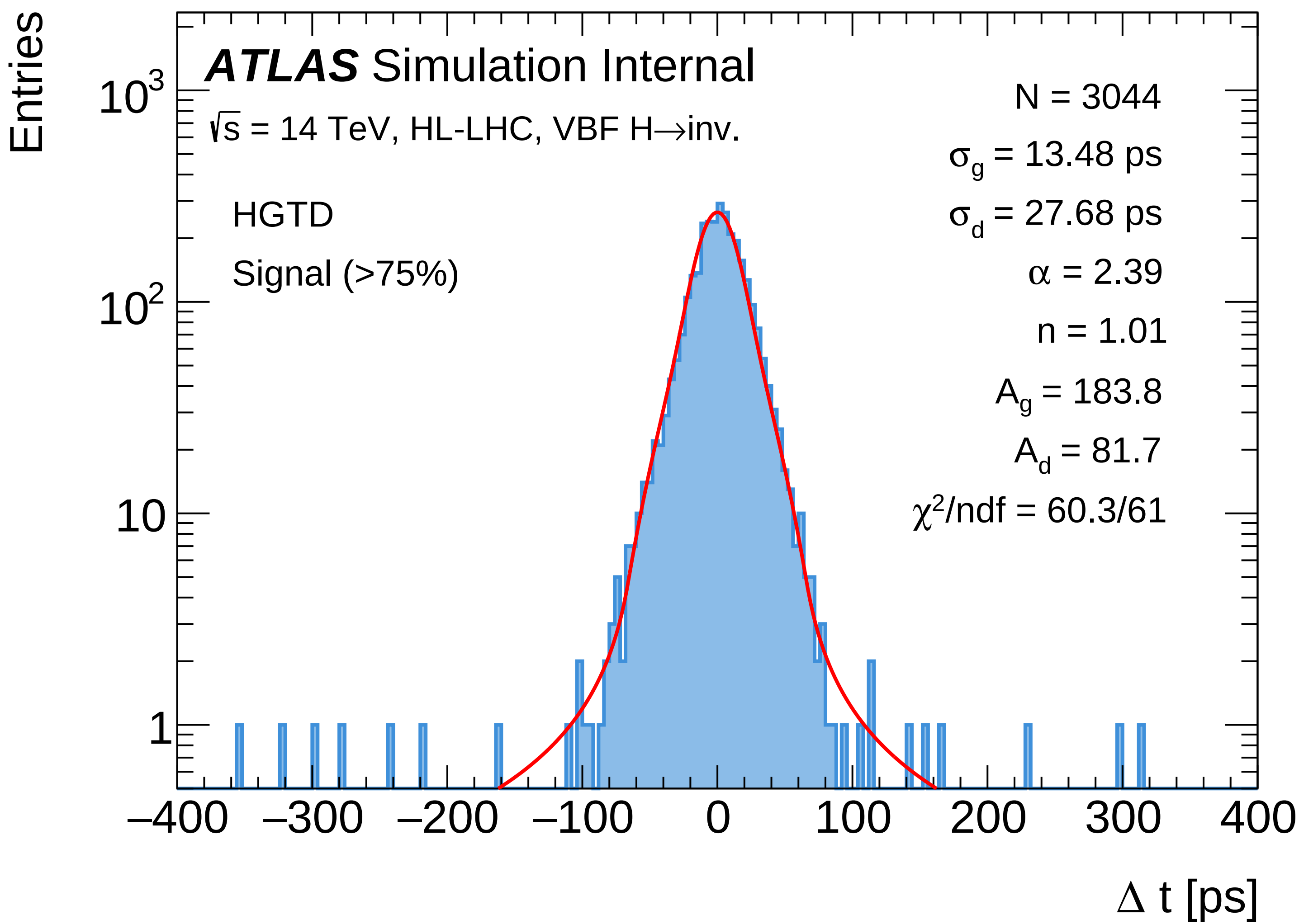
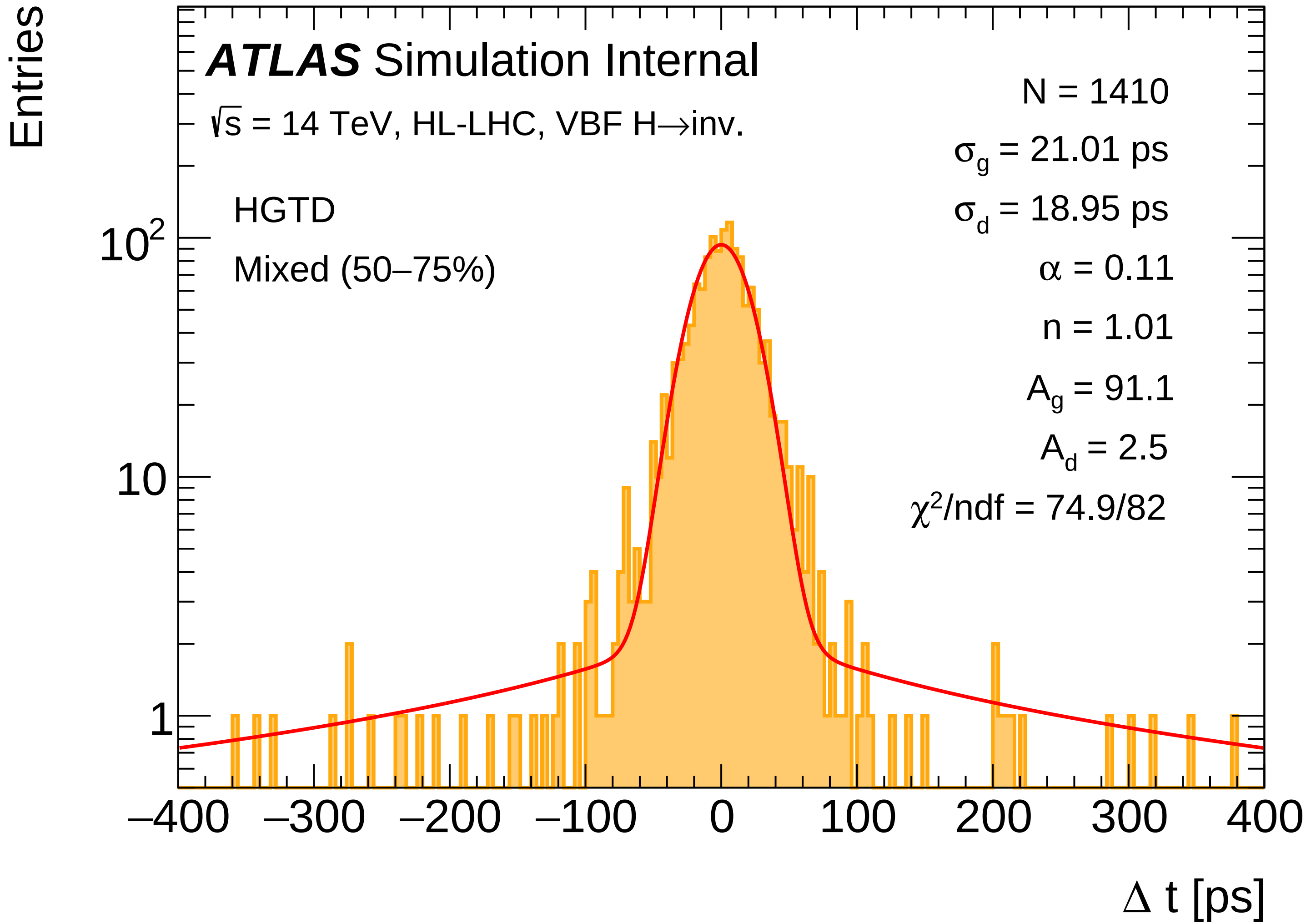
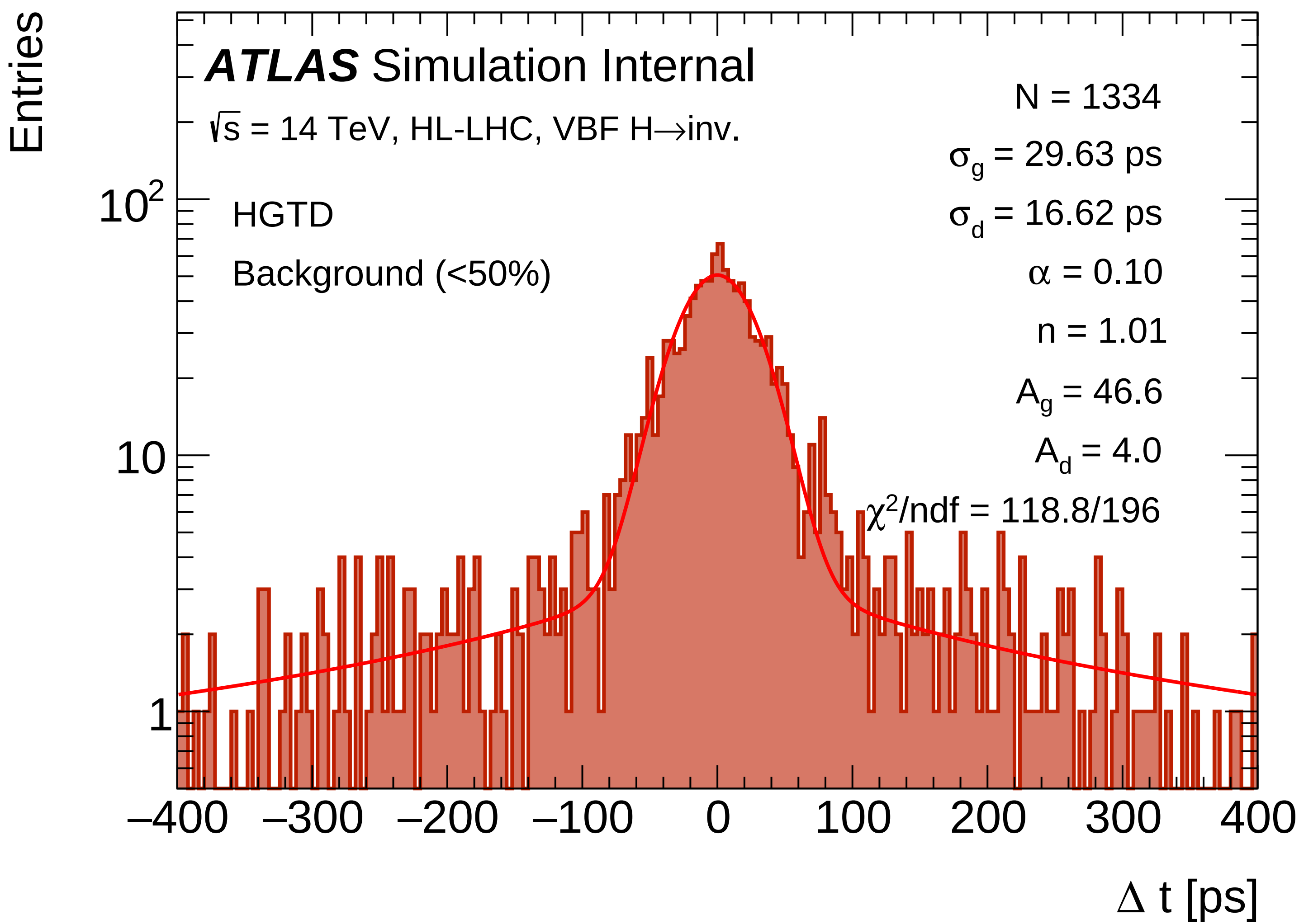


ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
HGTD
σg=12.93, σd=29.92
α=2.08, n=1.01
χ2/ndf=172.9/212
(S+M)/B=76.40
Signal (>75%)
Mixed (50–75%)
Background (<50%)
Gauss+DSCB Fit
Entries
Δ t [ps]

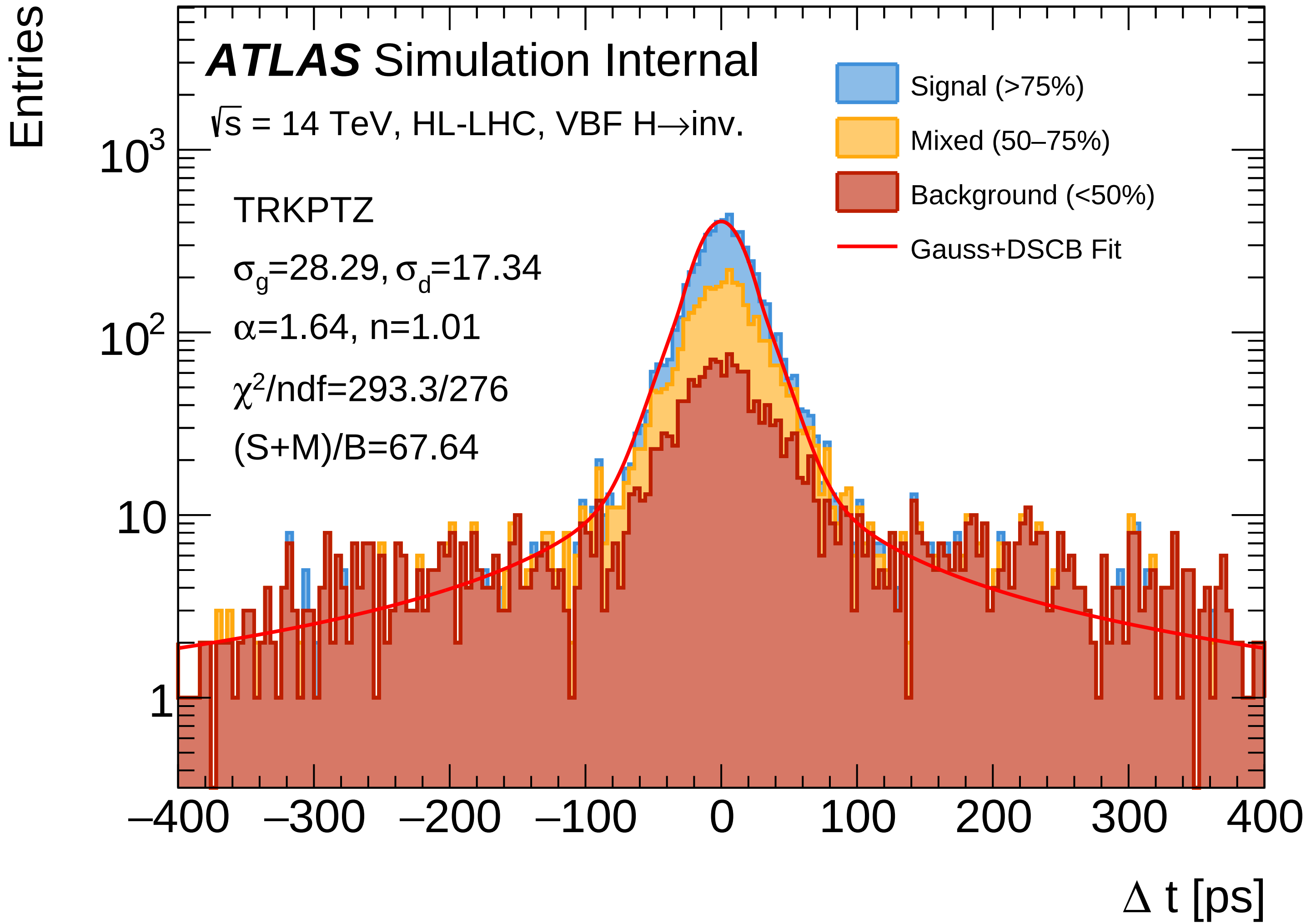


ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
HGTD
Signal (>75%)
N = 3044
σ_g = 13.48 ps
σ_d = 27.68 ps
α = 2.39
n = 1.01
A_g = 183.8
A_d = 81.7
χ²/ndf = 60.3/61
Entries
Δ t [ps]

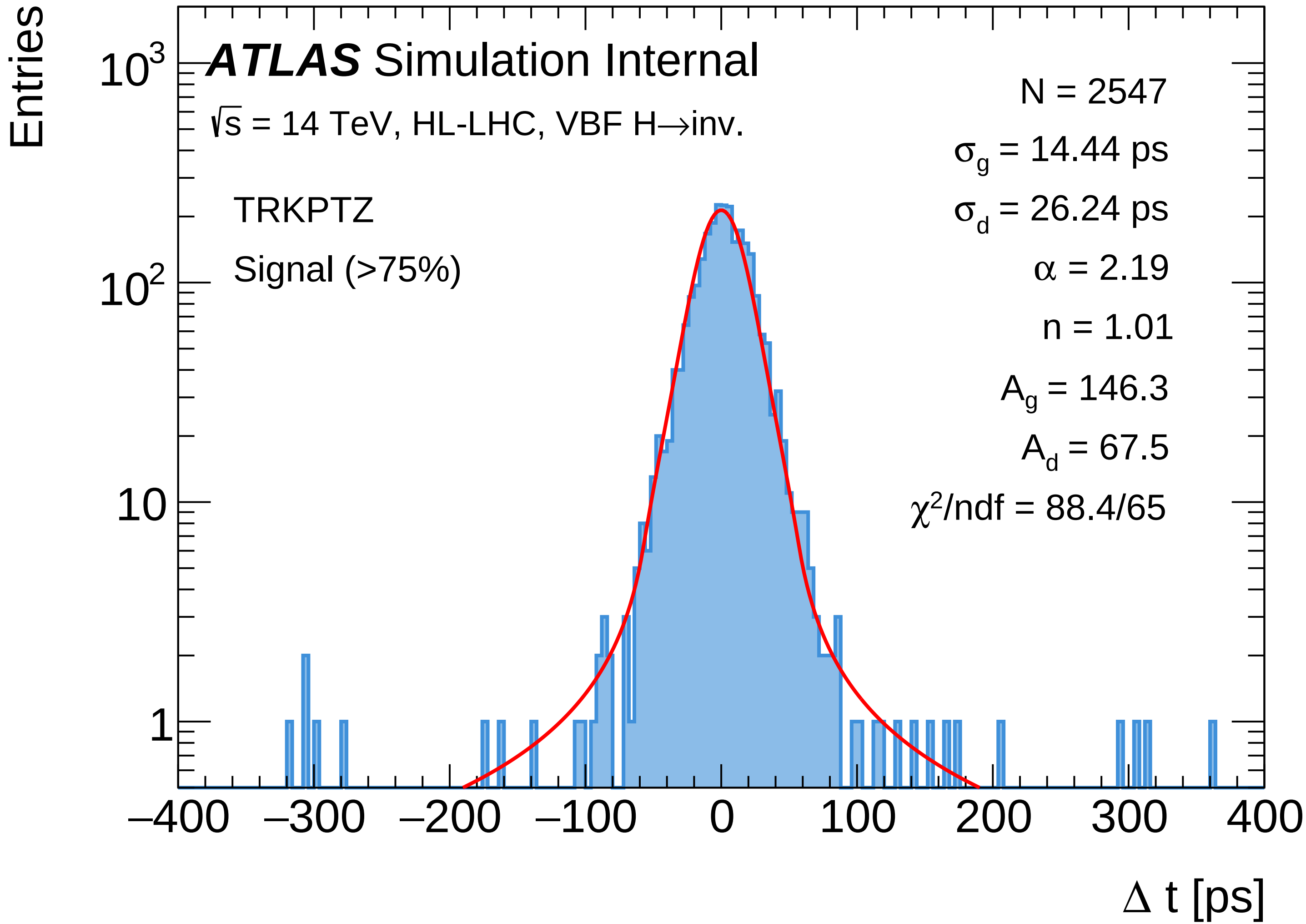

ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
HGTD
Mixed (50–75%)
N = 1410
σg = 21.01 ps
σd = 18.95 ps
α = 0.11
n = 1.01
Ag = 91.1
Ad = 2.5
χ2/ndf = 74.9/82
Entries
Δ t [ps]
1
10
10^2
−400
−300
−200
−100
0
100
200
300
400

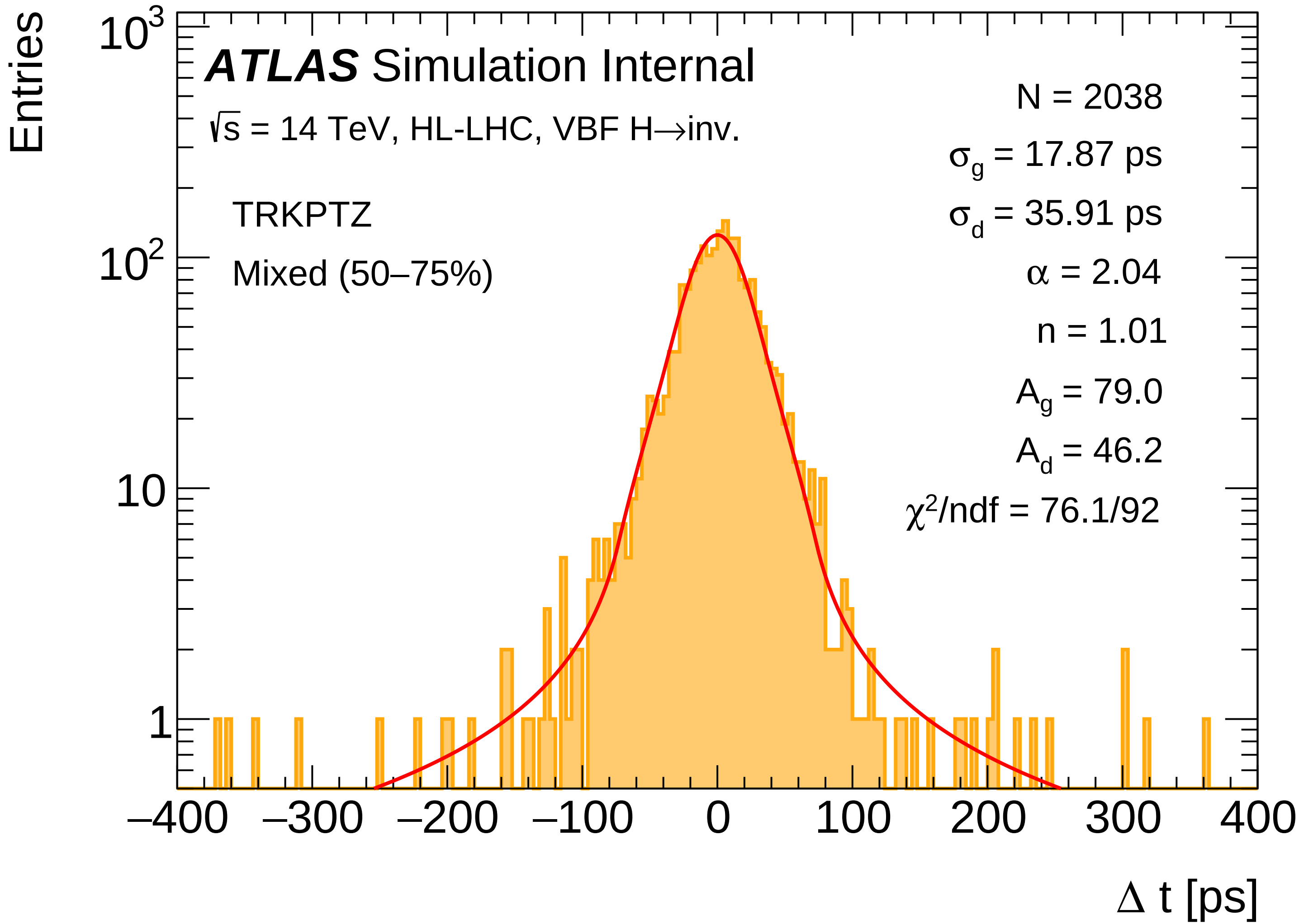


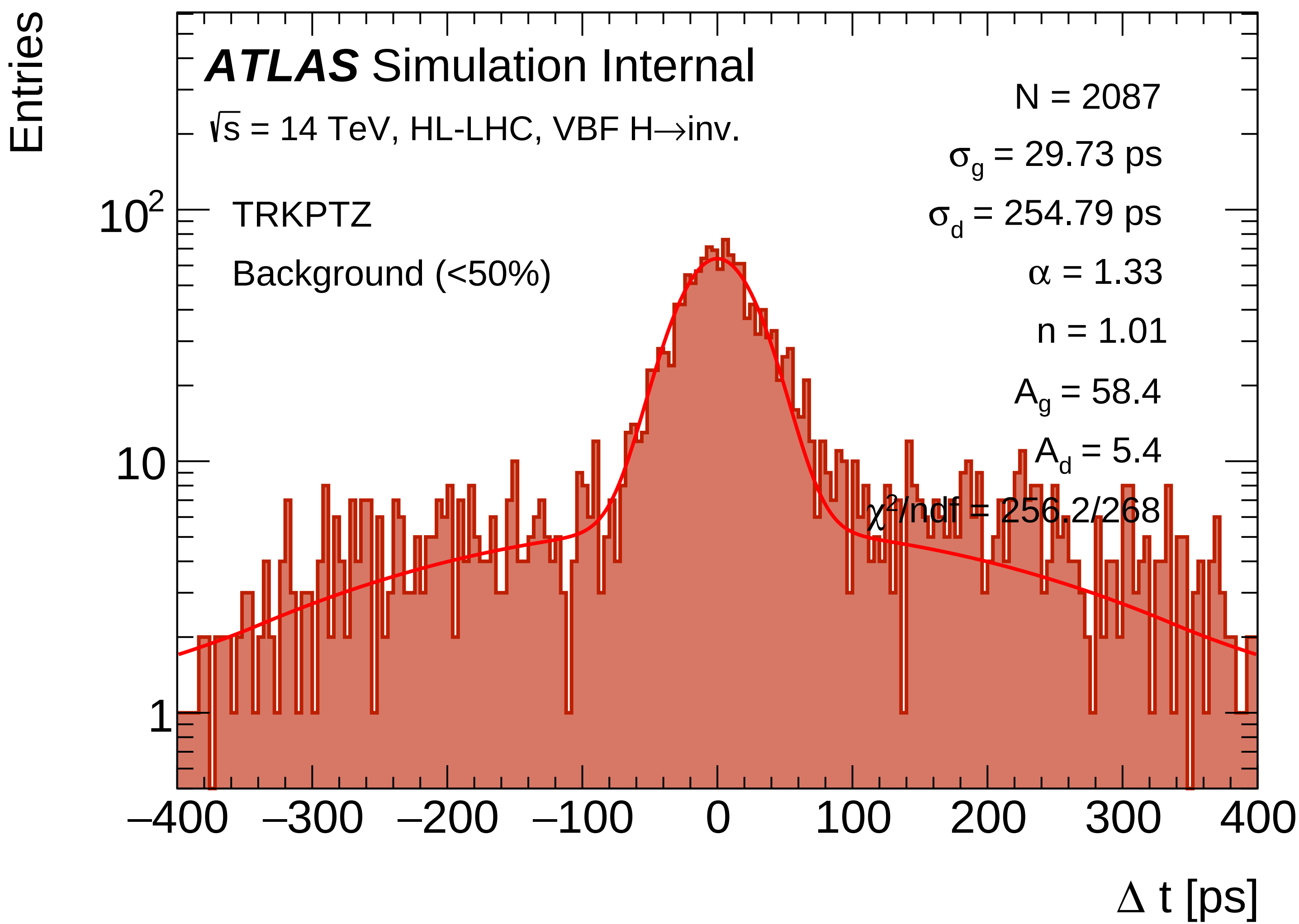
ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
HGTD
Background (<50%)
N = 1334
σ_g = 29.63 ps
σ_d = 16.62 ps
α = 0.10
n = 1.01
A_g = 46.6
A_d = 4.0
χ²/ndf = 118.8/196
Entries
Δ t [ps]



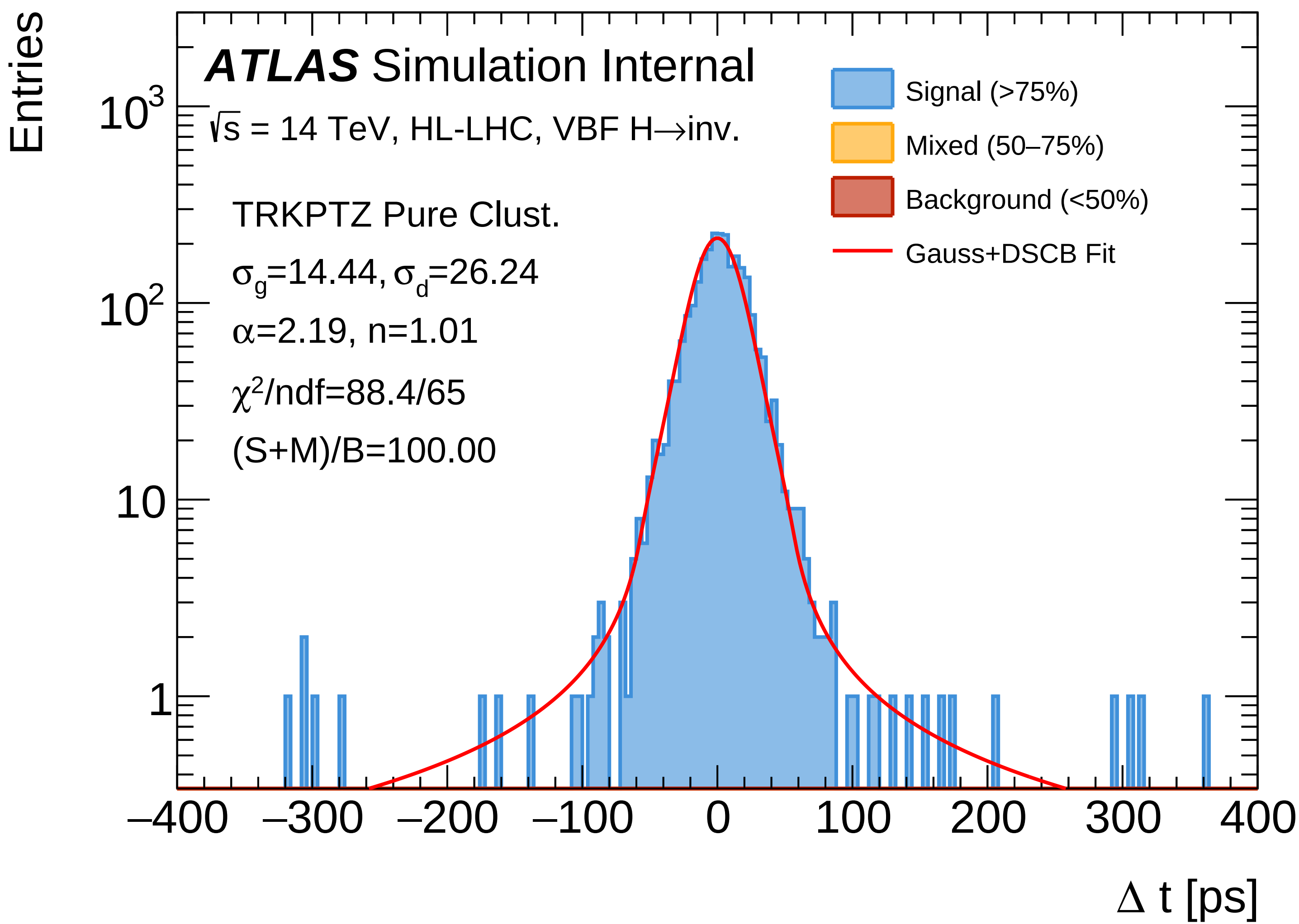
ATLAS Simulation Internal
$\sqrt{s}$ = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ
σ_g=28.29, σ_d=17.34
α=1.64, n=1.01
χ^2/ndf=293.3/276
(S+M)/B=67.64
Signal (>75%)
Mixed (50–75%)
Background (<50%)
Gauss+DSCB Fit
Entries
Δ t [ps]


ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ
Signal (>75%)
N = 2547
σg = 14.44 ps
σd = 26.24 ps
α = 2.19
n = 1.01
Ag = 146.3
Ad = 67.5
χ2/ndf = 88.4/65
Entries
Δ t [ps]
10^3
10^2
10
1
−400
−300
−200
−100
0
100
200
300
400

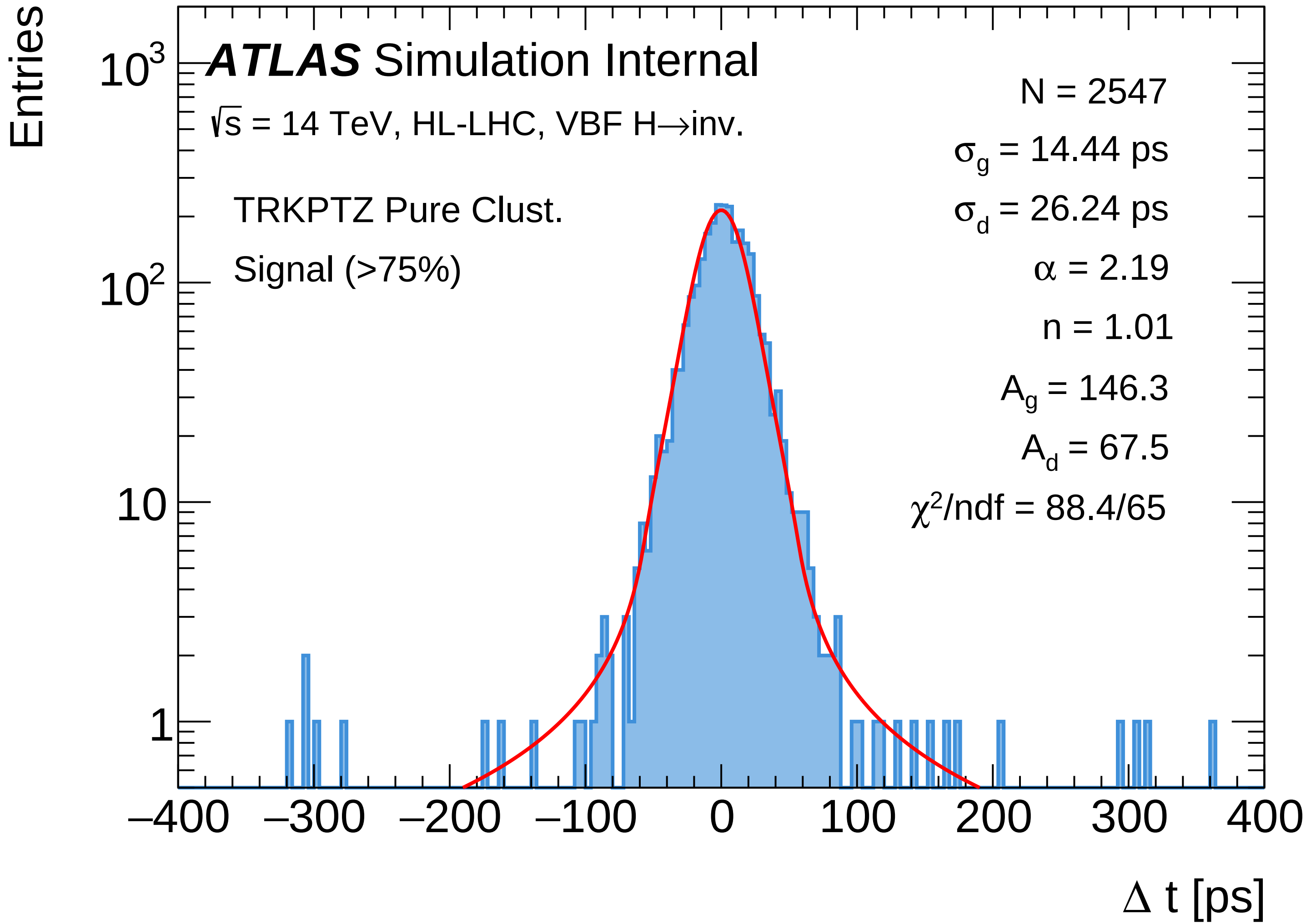

ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ
Mixed (50–75%)
N = 2038
σ_g = 17.87 ps
σ_d = 35.91 ps
α = 2.04
n = 1.01
A_g = 79.0
A_d = 46.2
χ²/ndf = 76.1/92
Entries
Δ t [ps]



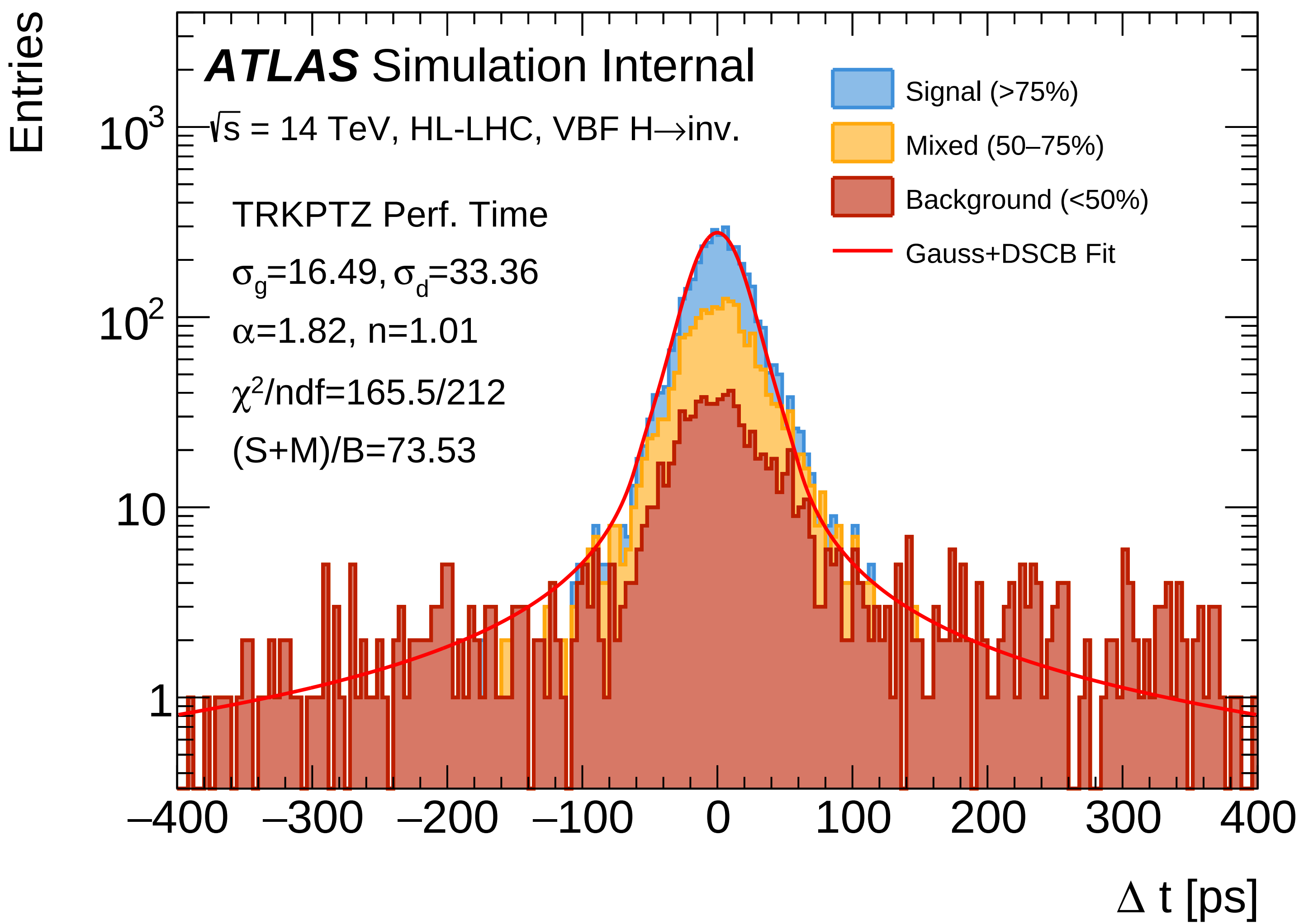
ATLAS Simulation Internal
$\sqrt{s}$ = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ
Background (<50%)
N = 2087
σ_g = 29.73 ps
σ_d = 254.79 ps
α = 1.33
n = 1.01
A_g = 58.4
A_d = 5.4
χ^2/ndf = 256.2/268
Entries
10^2
10
1
–400
–300
–200
–100
0
100
200
300
400
Δ t [ps]



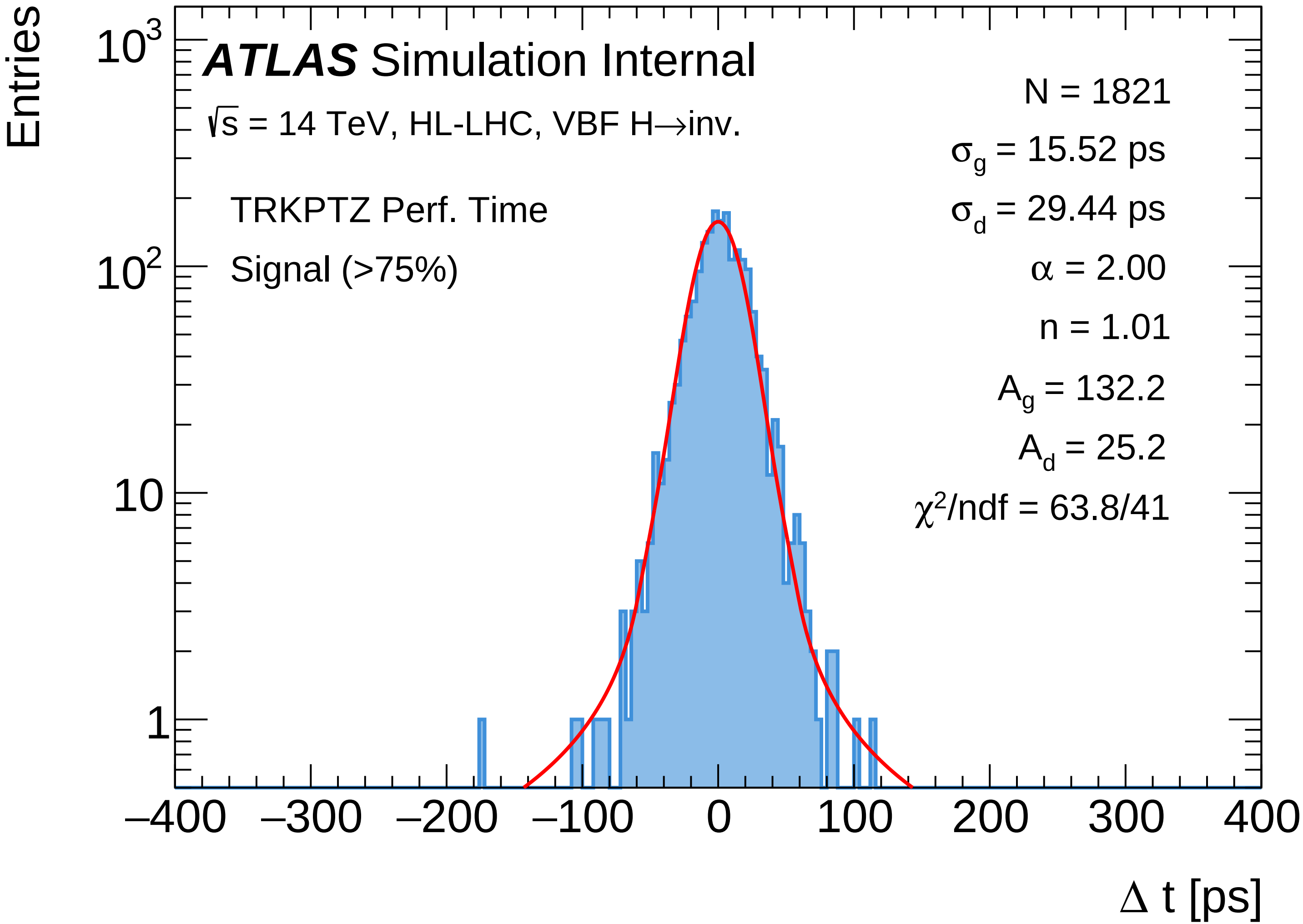
ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ Pure Clust.
σg=14.44, σd=26.24
α=2.19, n=1.01
χ2/ndf=88.4/65
(S+M)/B=100.00
Signal (>75%)
Mixed (50–75%)
Background (<50%)
Gauss+DSCB Fit
Entries
Δ t [ps]



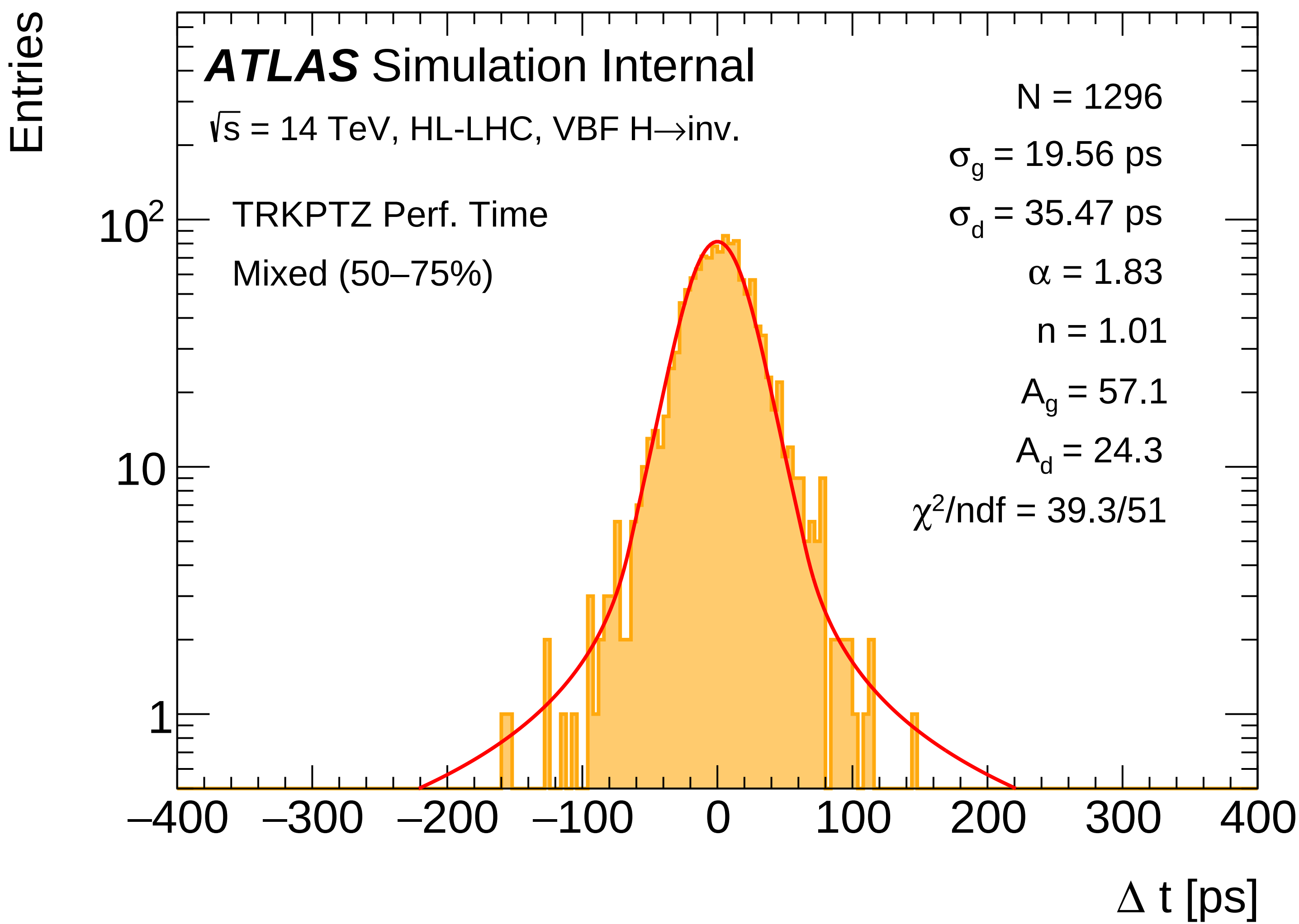
ATLAS Simulation Internal
$\sqrt{s}$ = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ Pure Clust.
Signal (>75%)
N = 2547
σ_g = 14.44 ps
σ_d = 26.24 ps
α = 2.19
n = 1.01
A_g = 146.3
A_d = 67.5
χ^2/ndf = 88.4/65
Entries
Δ t [ps]
1
10
10^2
10^3
−400
−300
−200
−100
0
100
200
300
400

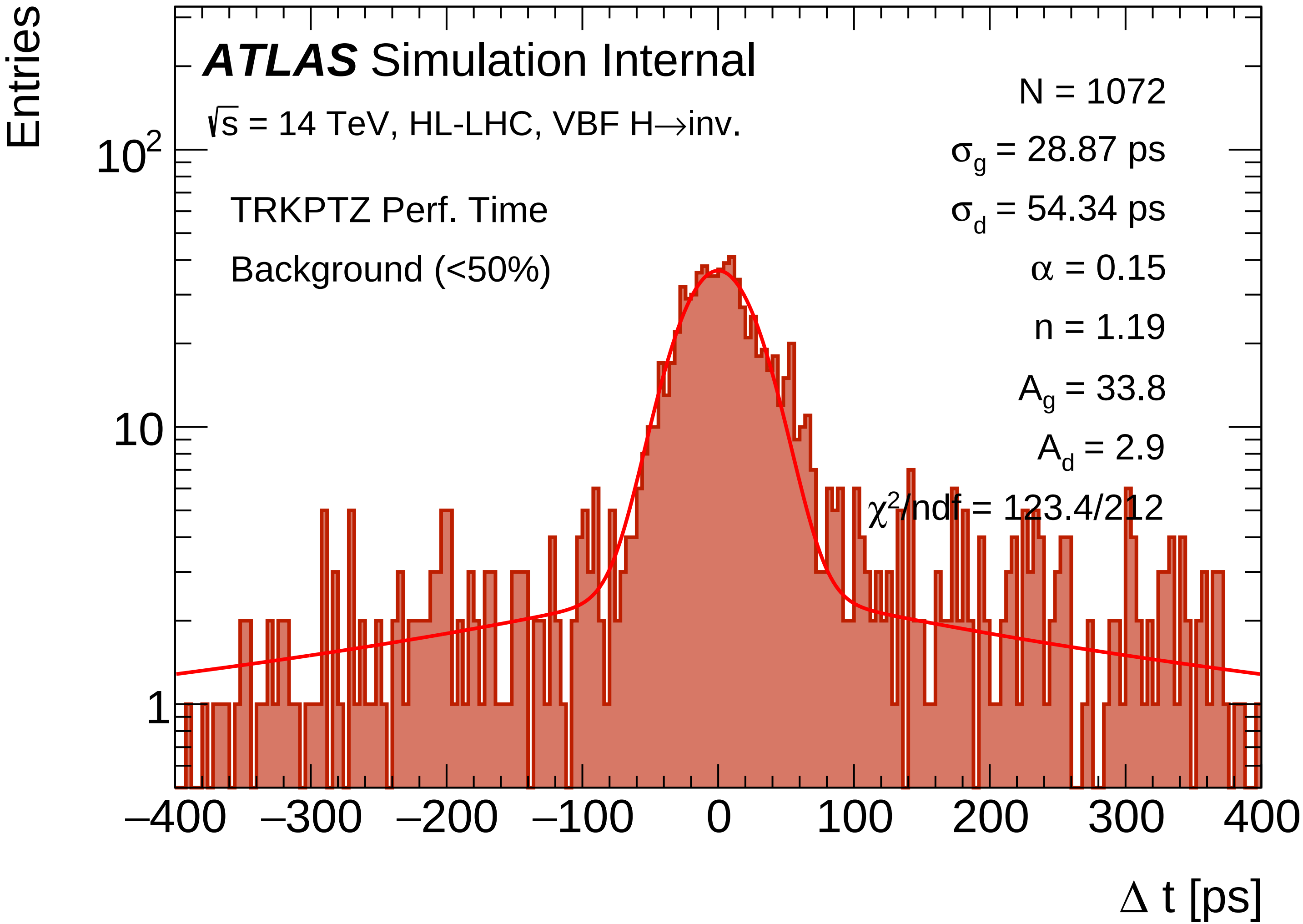


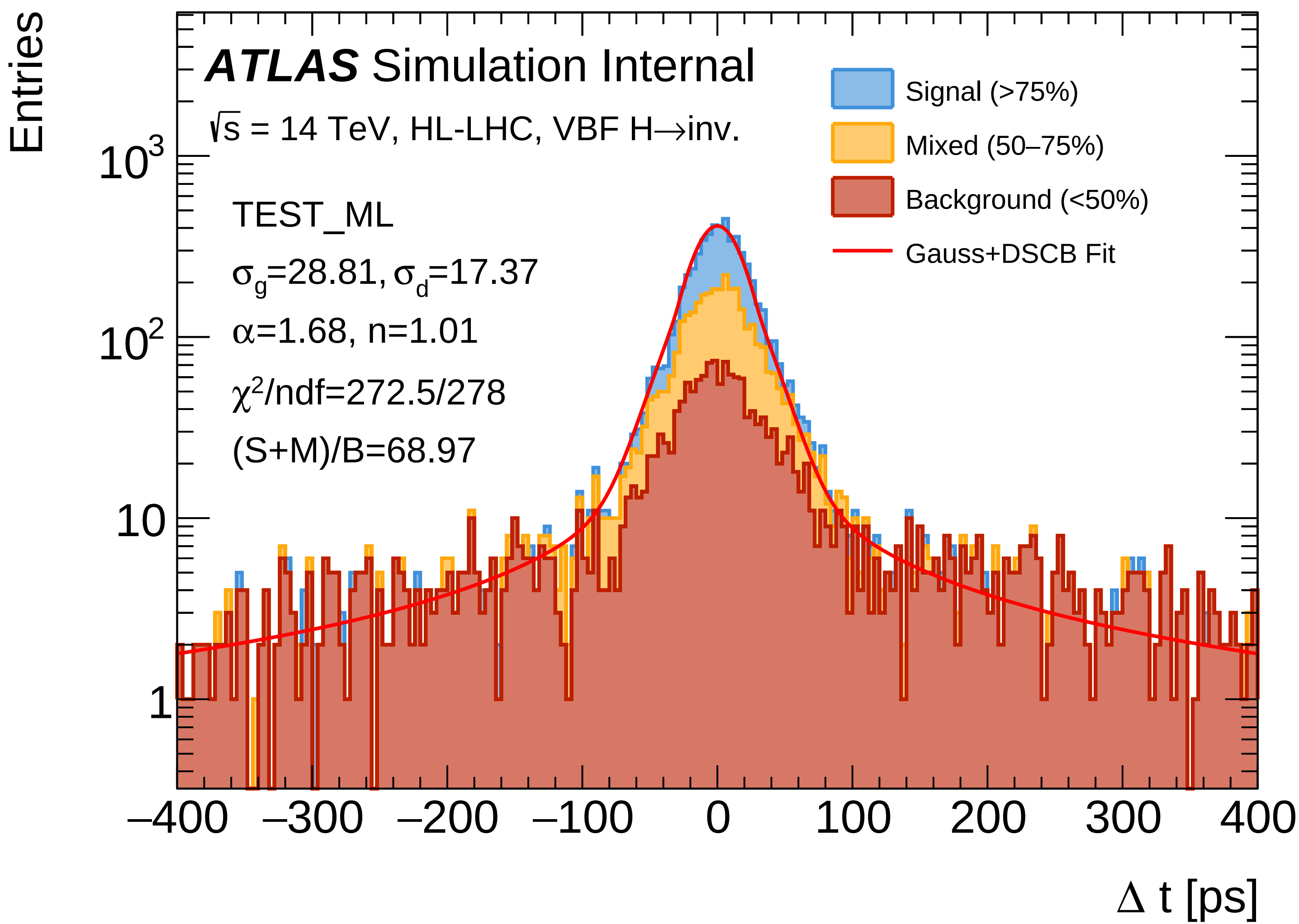
ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ Perf. Time
σg=16.49, σd=33.36
α=1.82, n=1.01
χ2/ndf=165.5/212
(S+M)/B=73.53
Signal (>75%)
Mixed (50–75%)
Background (<50%)
Gauss+DSCB Fit
Entries
1
10
10^2
10^3
–400
–300
–200
–100
0
100
200
300
400
Δ t [ps]



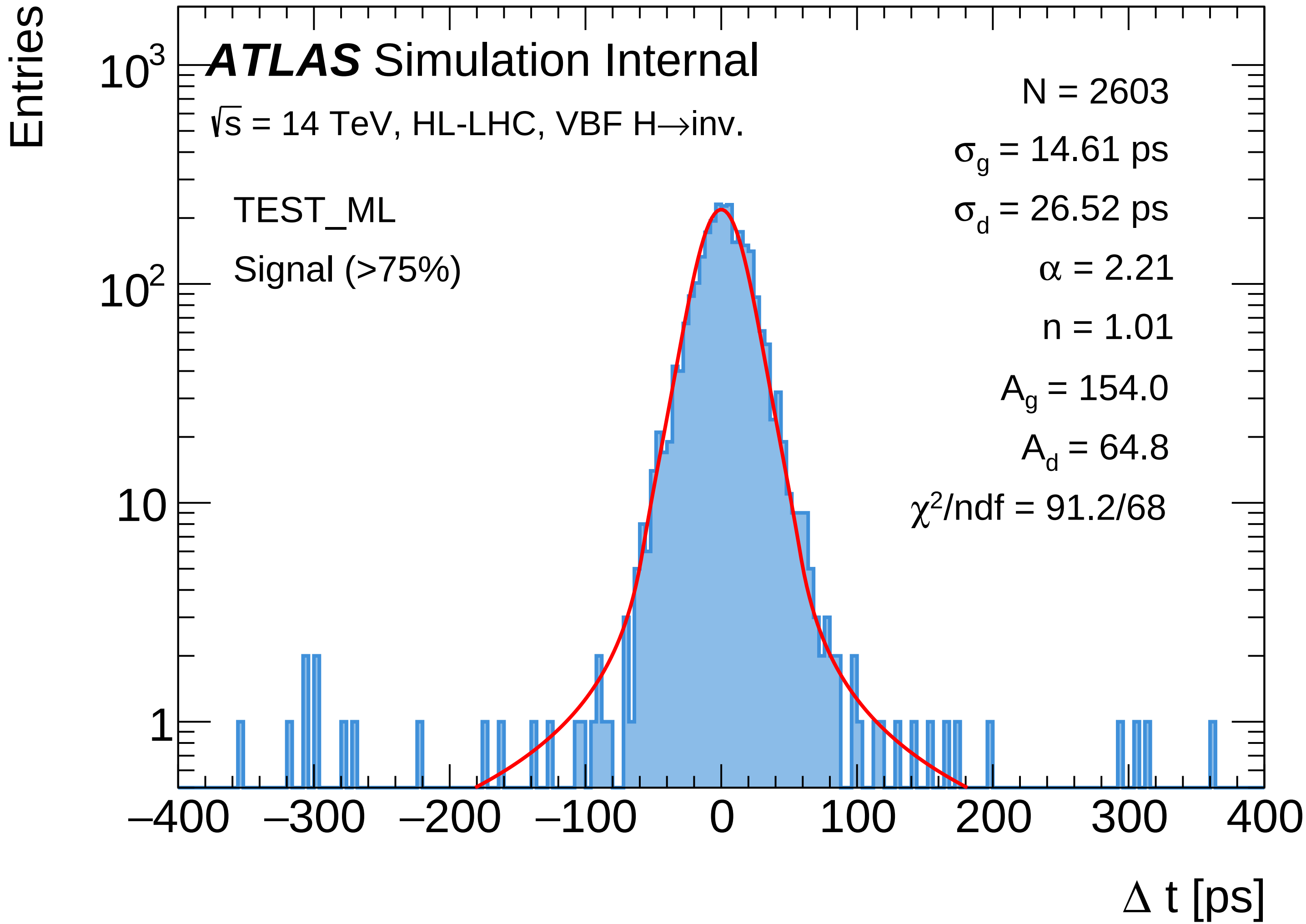
ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ Perf. Time
Signal (>75%)
N = 1821
σg = 15.52 ps
σd = 29.44 ps
α = 2.00
n = 1.01
Ag = 132.2
Ad = 25.2
χ2/ndf = 63.8/41
Entries
10^3
10^2
10
1
−400
−300
−200
−100
0
100
200
300
400
Δ t [ps]


ATLAS Simulation Internal
$\sqrt{s}$ = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ Perf. Time
Mixed (50–75%)
N = 1296
σ_g = 19.56 ps
σ_d = 35.47 ps
α = 1.83
n = 1.01
A_g = 57.1
A_d = 24.3
χ^2/ndf = 39.3/51
Entries
10^2
10
1
–400
–300
–200
–100
0
100
200
300
400
Δ t [ps]

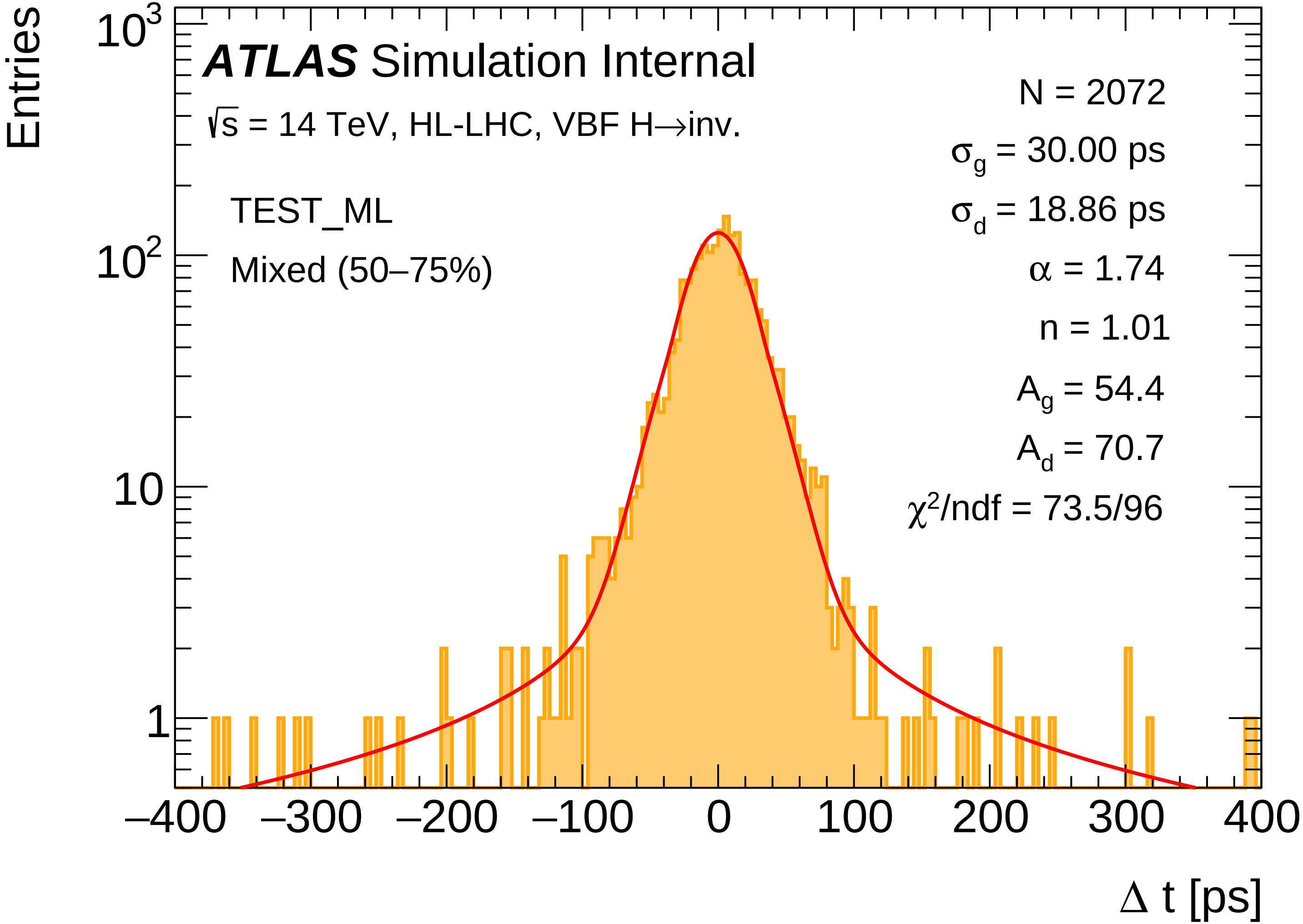

ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TRKPTZ Perf. Time
Background (<50%)
N = 1072
σg = 28.87 ps
σd = 54.34 ps
α = 0.15
n = 1.19
Ag = 33.8
Ad = 2.9
χ2/ndf = 123.4/212
Entries
Δ t [ps]
10^2
10
1
–400
–300
–200
–100
0
100
200
300
400

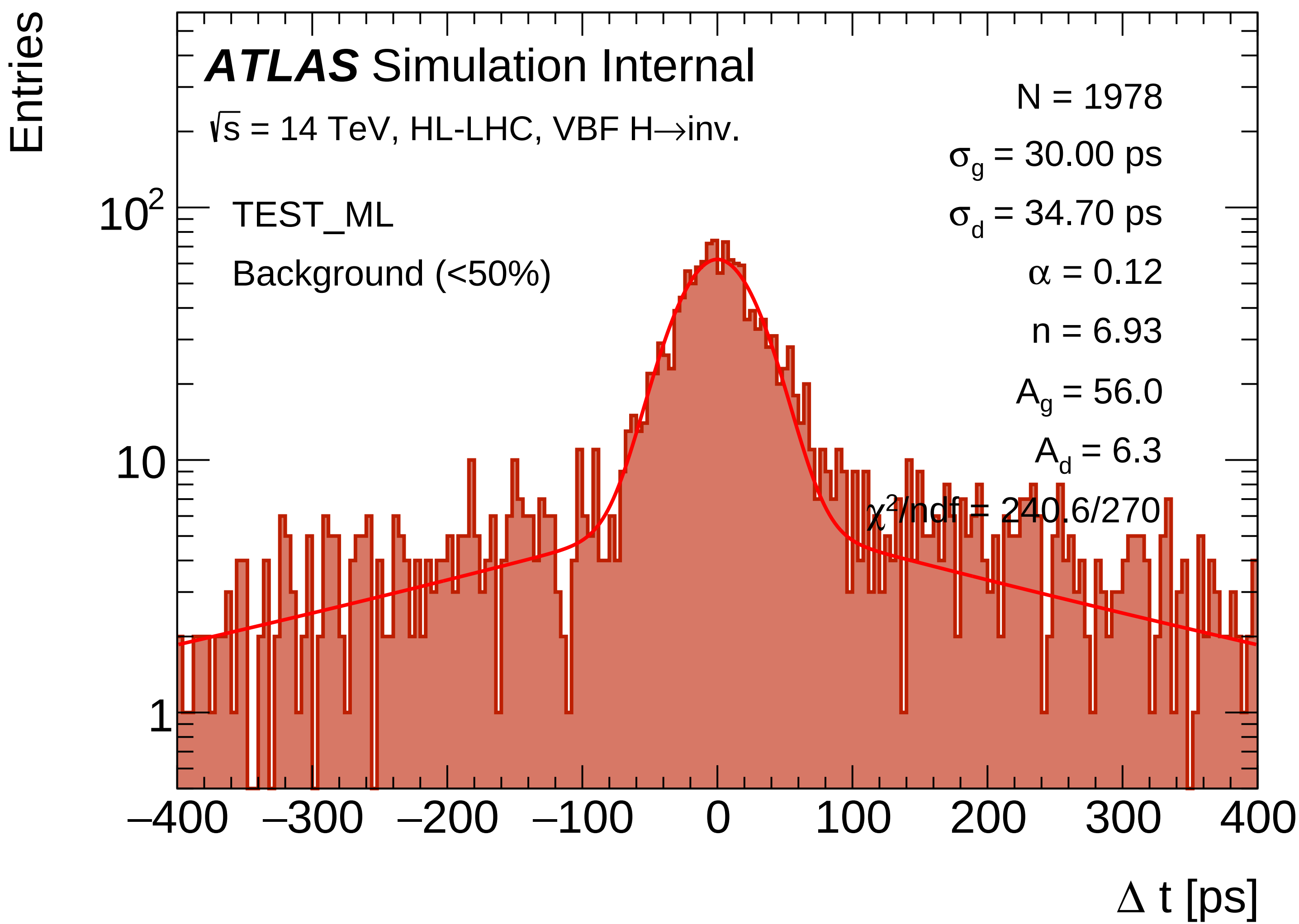


ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TEST_ML
σ_g=28.81, σ_d=17.37
α=1.68, n=1.01
χ²/ndf=272.5/278
(S+M)/B=68.97
Signal (>75%)
Mixed (50–75%)
Background (<50%)
Gauss+DSCB Fit
Entries
Δ t [ps]
10³
10²
10
1
–400
–300
–200
–100
0
100
200
300
400



ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TEST_ML
Signal (>75%)
N = 2603
σg = 14.61 ps
σd = 26.52 ps
α = 2.21
n = 1.01
Ag = 154.0
Ad = 64.8
χ2/ndf = 91.2/68
Entries
Δ t [ps]


ATLAS Simulation Internal
√s = 14 TeV, HL-LHC, VBF H→inv.
TEST_ML
Mixed (50–75%)
N = 2072
σ_g = 30.00 ps
σ_d = 18.86 ps
α = 1.74
n = 1.01
A_g = 54.4
A_d = 70.7
χ²/ndf = 73.5/96
Entries
Δ t [ps]
10³
10²
10
1
−400
−300
−200
−100
0
100
200
300
400


ATLAS Simulation Internal
$\sqrt{s}$ = 14 TeV, HL-LHC, VBF H→inv.
TEST_ML
Background (<50%)
N = 1978
σ_g = 30.00 ps
σ_d = 34.70 ps
α = 0.12
n = 6.93
A_g = 56.0
A_d = 6.3
χ^2/ndf = 240.6/270
Entries
10^2
10
1
−400
−300
−200
−100
0
100
200
300
400
Δ t [ps]